

# Uniqueness of the Kähler structure of $\mathbb{CP}^n$

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# 1 Overview

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- In this talk we mainly focus on the following two theorems, which show the uniqueness of the Kähler structure of  $\mathbb{CP}^n$  in different categories.

### Theorem (Hirzebruch, Kodaira, 1957; Yau, 1977)

*If a Kähler manifold  $M$  is homeomorphic to  $\mathbb{CP}^n$ , then  $M$  is biholomorphic to it.*

### Theorem (Yau, 1977)

*If a compact complex surface  $M$  is homotopy equivalent to  $\mathbb{CP}^2$ , then  $M$  is biholomorphic to it.*

- To prove these two theorems, the following lemma motivates us it suffices to construct a holomorphic line bundle with some properties.

### Lemma (Kobayashi, Ochiai, 1973)

*If  $M$  is a compact Kähler  $n$ -manifold and  $L$  is a positive holomorphic line bundle over  $M$  with  $\int_M c_1(L)^n = 1$  and  $\dim H^0(M, L) = n + 1$ , then  $M$  is biholomorphic to  $\mathbb{CP}^n$ .*

# Rough idea of proof

- If  $M$  is a compact Kähler manifold whose cohomology groups are the same as the ones of  $\mathbb{CP}^n$ , then
  - $c_1: \text{Pic}(M) \rightarrow H^2(M, \mathbb{Z})$  is an isomorphism, which allows us to construct a (unique) holomorphic line bundle  $L$  with a given cohomology class as its first Chern class.

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- By using Kodaira vanishing theorem one can conclude  $H^k(M, L) = 0$  for  $k > 0$ . In particular, one has  $\dim H^0(M, L) = n + 1$ , as desired.

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- Choose a Kähler form  $\tilde{\omega}$  on  $M$ . Its cohomology class lies in  $H^2(M, \mathbb{R}) \cong \mathbb{R}$ , so we can rescale  $\tilde{\omega}$  to get another Kähler form  $\omega$  whose cohomology class generates  $H^2(M, \mathbb{Z}) \cong \mathbb{Z}$ . In particular,  $\int_M \omega^n = 1$ .

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- Since  $c_1$  is an isomorphism, there exists a (unique) holomorphic line bundle  $L$  whose first Chern class is  $[\omega]$ .

## Lemma

*For any holomorphic line bundle  $L$  over  $M$  we have*

$$\chi(M, L) = \int_M e^{c_1(L) + \frac{c_1(M)}{2}} \left( \frac{\omega/2}{\sinh(\omega/2)} \right)^{n+1}.$$

## Corollary

*$c_1(M)$  equals either  $(n+1)[\omega]$  or  $-(n+1)[\omega]$ , with the latter only possibly occurring when  $n$  is even.*

## Proof.

Since  $[\omega]$  is a generator of  $H^2(M, \mathbb{Z})$ , we may write  $c_1(M) = \lambda[\omega]$ . The reduction mod 2 of  $c_1(M)$  is the second Stiefel-Whitney class  $w_2(M) \in H^2(M, \mathbb{Z})$ , which is a topological invariant. Hence it is equal to  $w_2(\mathbb{CP}^n)$  which equals  $c_1(\mathbb{CP}^n) \equiv n+1 \pmod{2}$ . This shows  $c_1(M) = (n+1+2s)[\omega]$  for some  $s \in \mathbb{Z}$ .

## Continuation.

By Lemma 4 one has

$$\chi(M, \mathcal{O}) = \int_M e^{\frac{n+1+2s}{2}\omega} \left( \frac{\omega/2}{\sinh(\omega/2)} \right)^{n+1} = \int_M e^{s\omega} \left( \frac{\omega}{1 - e^{-\omega}} \right)^{n+1}.$$

By residue theorem a direct computation shows

$$\int_M e^{s\omega} \left( \frac{\omega}{1 - e^{-\omega}} \right)^{n+1} = \binom{n+s}{n}.$$

Since  $\chi(M, \mathcal{O}) = 1$ , one has  $\binom{n+s}{n} = 1$ , which can be rewritten as

$$n! = (s+n) \dots (s+1).$$

So if  $n$  is odd this implies  $s = 0$ , while if  $n$  is even,  $s$  is either 0 or  $-n-1$ . This completes the proof. □

## Proof of Theorem 1.

**Case I:** Assume first  $c_1(M) = (n+1)[\omega]$ , which implies that  $M$  is a Fano manifold. Then  $c_1(K_M) = -c_1(M) = -(n+1)c_1(L)$  and so  $K_M = -(n+1)L$  since  $c_1$  is an isomorphism. Then Serre duality gives  $H^k(M, L) = H^{n-k}(M, K_M - L)$  and  $K_M - L = -(n+2)L$  is negative, so  $H^k(M, L) = 0$  if  $k > 0$  by Kodaira vanishing. Hence one has

$$\dim H^0(M, L) = \chi(M, L) = \int_M e^{c_1(L) + \frac{c_1(M)}{2}} \left( \frac{\omega/2}{\sinh(\omega/2)} \right)^{n+1} = n+1,$$

and Lemma 3 implies  $M$  is biholomorphic to  $\mathbb{CP}^n$ .

**Case II:** Assume  $c_1(M) = -(n+1)[\omega] < 0$ , it suffices to show the following identity

$$(2(n+1)c_2(M) - nc_1^2(M)) [\omega]^{n-2} = 0.$$



## Continuation.

Indeed, by the equality condition of Chern number inequality of Yau,  $M$  has constant holomorphic bisectional curvature  $-1$ , and thus by uniformization theorem  $M$  is biholomorphic to the unit ball in  $\mathbb{C}^n$ , a contradiction.

To compute  $c_2(M)$ , note that  $p_1(M) = p_1(TM) = -c_2(TM \otimes \mathbb{C})$ ,  $TM \otimes \mathbb{C} \cong TM \oplus \overline{TM}$  and Chern classes satisfy  $c_k(\overline{TM}) = (-1)^k c_k(TM)$ , so

$$\begin{aligned} p_1(M) &= -c_2(TM \oplus \overline{TM}) \\ &= -c_2(TM) - c_2(\overline{TM}) - c_1(TM)c_1(\overline{TM}) \\ &= -2c_2(M) + c_1^2(M). \end{aligned}$$

On the other hand,  $p_1(M) = (n+1)[\omega]^2$ . Thus

$$2c_2(M) = (n+1)^2[\omega]^2 - (n+1)[\omega]^2 = n(n+1)[\omega]^2.$$

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## Proof of Theorem 2.

Let  $\tau(M)$  denote the signature of  $M$ , that is the signature of its intersection form. The signature is a topological invariant up to sign, and so

$$\tau(M) = \pm \tau(\mathbb{CP}^2) = \pm 1.$$

Hirzebruch's signature theorem gives

$$\tau(M) = \frac{1}{3} \int_M p_1(M) = \frac{1}{3} \int_M (c_1^2(M) - 2c_2(M)) = \pm 1.$$

Chern-Gauss-Bonnet's theorem gives

$$\int_M c_2(M) = \chi(M) = \chi(\mathbb{CP}^2) = 3.$$

As a consequence,  $\int_M c_1^2(M) \neq 0$ , which implies  $M$  is Kähler by Kodaira embedding.

## Continuation.

As before we see that  $\chi(M, \mathcal{O}) = 1$  and Hirzebruch-Riemann-Roch gives

$$\chi(M, \mathcal{O}) = \frac{K_M^2 + \chi(M)}{12} = \frac{K_M^2 + 3}{12},$$

which gives  $\int_M c_1^2(M) = K_M^2 = 9$ . Let  $\omega$  be as before, and  $c_1(M) = \lambda[\omega]$  for some  $\lambda \in \mathbb{Z}$ . Then  $\lambda = \pm 3$ . Here it suffices to show in case  $\lambda = 3$ ,  $\dim H^0(M, L) = 3$ , and the case  $\lambda = -3$  leads the same contradiction as before. By Hirzebruch-Riemann-Roch formula one has

$$\chi(M, L) = 1 + \frac{L^2 - K_M \cdot L}{2} = 3.$$

Serre duality and Kodaira vanishing gives

$H^1(M, L) = H^2(M, L) = 0$  as before. So

$\dim H^0(M, L) = \chi(M, L) = 3$ . This completes the proof.



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- A natural question is that whether the Kähler hypothesis is really necessary in Theorem 1. If not, one can also ask whether a complex manifold diffeomorphic to  $\mathbb{CP}^n$  must be biholomorphic to it. If it's true when  $n = 3$ , then there is no complex structure on  $S^6$ .

## Lemma

*If there exists a compact complex manifold  $M$  diffeomorphic to  $S^6$ , then there exists a compact complex manifold  $\tilde{M}$  diffeomorphic to  $\mathbb{CP}^3$  but not biholomorphic to it.*

## Proof.

Let  $M$  be a compact complex manifold diffeomorphic to  $S^6$  and  $\tilde{M}$  be its blowup at one point  $p \in M$ . A basic fact is that  $\tilde{M}$  is a compact complex manifold which is diffeomorphic to  $S^6 \# \overline{\mathbb{CP}^3}$ , where  $\overline{\mathbb{CP}^3}$  is the smooth manifold obtained from  $\mathbb{CP}^3$  by reversing orientation. In particular,  $\tilde{M}$  is diffeomorphic to  $\mathbb{CP}^3$ . If  $\tilde{M}$  was biholomorphic to  $\mathbb{CP}^3$ , one has

$$\int_{\tilde{M}} c_1(\tilde{M})^3 = \int_{\mathbb{CP}^3} c_1(\mathbb{CP}^3)^3 = 64$$



## Continuation.

On the other hand, if we let  $\pi: \tilde{M} \rightarrow M$  be the blow up map and  $E = \pi^{-1}(p)$  be its exceptional divisor, then one has

$$c_1(\tilde{M}) = \pi^* c_1(M) - 2[E]$$

where  $[E]$  is the Poincaré duality of  $E$ . Since  $b_2(M) = 0$ , one has  $c_1(M) = 0$ . Thus

$$\begin{aligned} \int_{\tilde{M}} c_1(\tilde{M})^3 &= -8 \int_{\tilde{M}} [E]^3 \\ &= -8 \int_E [E]^2 \\ &= -8 \int_{\mathbb{CP}^2} c_1(\mathcal{O}(-1))^2 = -8 \end{aligned}$$

Therefore  $\tilde{M}$  is not biholomorphic to  $\mathbb{CP}^3$ , as desired.

*Thanks!*